CS59300-CVD: Computer Vision with Deep Learning
Spring 2026
Midterm Exam — Questions & Solutions

Exam Rules:

1. This exam is **closed book, closed notes**. No electronic devices are permitted.
2. You have **75 minutes** to complete the exam.
3. There are **60 points** total across 5 sections.
4. Write your answers clearly in the provided answer boxes.
5. For multiple choice questions, write your letter answer in the designated box.
6. Show your work where requested. Partial credit may be awarded.
7. If you need extra space, use the back of the page and indicate clearly.

Point Summary:

Section	**Points**
1. Deep Learning & Differentiable Operations	10
2. Image Processing	10
3. Fitting & Alignment	10
4. Short Answer	22
5. Multiple Choice	8
Total	**60**

1. [10 points] Deep Learning & Differentiable Operations

(a) [**4 points**] Consider the following PyTorch code:

```
import torch
import torch.nn as nn
A = torch.randn(10, 3)
c = torch.randn(3)
b = torch.matmul(A, c)
w = torch.zeros(3, requires_grad=True)
optimizer = torch.optim.SGD([w], lr=0.01)
for step in range(1000):
    loss = torch.norm(torch.matmul(A, w) - b) ** 2
    loss.backward()
    optimizer.step()
```

(i) What optimization problem is this code attempting to solve? Write it explicitly in mathematical form. What is the optimal solution $w^\star$?

Solution

The code solves the least-squares problem:

$$\min_{\mathbf{w}} \|\mathbf{A}\mathbf{w} - \mathbf{b}\|^2$$

where $\mathbf{A} \in \mathbb{R}^{10\times 3}$ and $\mathbf{b} = \mathbf{A}\mathbf{c}$.

Since $\mathbf{b} = \mathbf{A}\mathbf{c}$, the vector $\mathbf{b}$ lies in the column space of $\mathbf{A}$, so the system is consistent. The unique least-squares solution is:

$$\boxed{\mathbf{w}^\star = \mathbf{c}}$$

(ii) The code above contains a bug. Identify the bug, explain precisely what it causes, and describe how to fix it.

Solution

Bug: The gradients are never zeroed between iterations. There is no call to `optimizer.zero_grad()` before `loss.backward()`.

Effect: PyTorch *accumulates* gradients by default. Without zeroing, `w.grad` at step t contains $\sum_{k=0}^{t} \nabla_{\mathbf{w}} \mathcal{L}_k$, so the optimizer uses a corrupted (ever-growing) gradient. The optimization will not converge correctly.

Fix: Add `optimizer.zero_grad()` before `loss.backward()`:

```
for step in range(1000):
    optimizer.zero_grad()          # <-- fix
    loss = torch.norm(torch.matmul(A, w) - b) ** 2
    loss.backward()
    optimizer.step()
```

(b) **[4 points]** Recall differentiable image sampling using the bilinear sampling kernel is

$$V[x_i', y_i'] = \sum_n^H \sum_m^W U[n,m] \max(0, 1 - |x_i - m|) \max(0, 1 - |y_i - n|) \quad \forall i$$

Please derive $\dfrac{\partial V[x_i', y_i']}{\partial U[n,m]}$ and $\dfrac{\partial V[x_i', y_i']}{\partial x_i}$.

Solution

Derivative w.r.t. $U[n,m]$: $U[n,m]$ appears linearly, so:

$$\boxed{\frac{\partial V[x_i', y_i']}{\partial U[n,m]} = \max(0,\, 1 - |x_i - m|)\ \max(0,\, 1 - |y_i - n|)}$$

This is simply the bilinear interpolation weight for grid point (m, n).

Derivative w.r.t. x_i: We need $\frac{\partial}{\partial x_i} \max(0, 1 - |x_i - m|)$. For $|x_i - m| < 1$:

$$\frac{\partial}{\partial x_i} \max(0,\, 1 - |x_i - m|) = \begin{cases} +1 & \text{if } m - 1 < x_i < m, \\ -1 & \text{if } m < x_i < m + 1, \\ 0 & \text{otherwise,} \end{cases}$$

which can be written compactly as $-\operatorname{sign}(x_i - m) \cdot \mathbf{1}_{|x_i - m| < 1}$. Therefore:

$$\boxed{\frac{\partial V[x_i', y_i']}{\partial x_i} = \sum_n \sum_m U[n,m] \left[-\operatorname{sign}(x_i - m)\right] \mathbf{1}_{|x_i - m| < 1}\ \max(0, 1 - |y_i - n|)}$$

(c) **[2 points]** Consider the following 4×4 input. We first apply 2×2 max pooling with stride 2 (no padding), then a ReLU activation. What is the output? Show your work.

-3	-1	2	-4
-2	-5	-3	7
1	-6	4	0
-5	8	-2	-1

Solution

Step 1 — Max pool over each 2×2 block:

Top-left: $\max(-3, -1, -2, -5) = -1$
Top-right: $\max(2, -4, -3, 7) = 7$
Bottom-left: $\max(1, -6, -5, 8) = 8$
Bottom-right: $\max(4, 0, -2, -1) = 4$

$$\Longrightarrow \begin{bmatrix} -1 & 7 \\ 8 & 4 \end{bmatrix}$$

Step 2 — ReLU ($\max(0, \cdot)$ element-wise):

$$\boxed{\begin{bmatrix} 0 & 7 \\ 8 & 4 \end{bmatrix}}$$

2. [10 points] Image Processing

(a) **[4 points]** Consider the two 3×3 filters below:

$$\mathbf{h}_1 = \frac{1}{6}\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \qquad \mathbf{h}_2 = \frac{1}{6}\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}.$$

(i) Show that $\mathbf{h}_1$ is separable by writing it as $\mathbf{h}_1 = \mathbf{a}\,\mathbf{b}^\top$ for column vectors $\mathbf{a}, \mathbf{b}$.

Solution

$$\mathbf{h}_1 = \frac{1}{6}\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} = \underbrace{\frac{1}{6}\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}}_{\mathbf{a}} \underbrace{\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}^\top}_{\mathbf{b}^\top}$$

Verification: $\mathbf{ab}^\top = \frac{1}{6}[1,2,1]^\top[-1,0,1] = \frac{1}{6}\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} = \mathbf{h}_1$. ✓

Here $\mathbf{a}$ is a smoothing (binomial) kernel along y, and $\mathbf{b}$ is a finite-difference kernel along x. This is the Sobel-x filter.

(ii) If we convolve an image with $\mathbf{h}_1$ followed by $\mathbf{h}_2$, what classical differential operator does the combined result approximate? Justify briefly.

Solution

$\mathbf{h}_1$ approximates $\partial/\partial x$ (horizontal gradient with vertical smoothing).
$\mathbf{h}_2$ approximates $\partial/\partial y$ (vertical gradient with horizontal smoothing).
Convolving with $\mathbf{h}_1$ then $\mathbf{h}_2$ yields:

$$\mathbf{h}_2 * (\mathbf{h}_1 * I) \;\approx\; \frac{\partial}{\partial y}\left(\frac{\partial I}{\partial x}\right) = \boxed{\frac{\partial^2 I}{\partial y\,\partial x}}$$

This is the **mixed second partial derivative** (cross-derivative), one of the off-diagonal terms of the **Hessian** of the image.

(b) **[2 points]** Design a 5×5 filter $\mathbf{h}$ that **shifts an image to the right by 2 pixels** when performing a **convolution**.

Solution

Under convolution the kernel is flipped, so to shift the output *right* by 2, we place a single 1 two columns to the *left* of center (center = row 2, col 2; impulse at row 2, col 0, 0-indexed):

$$\mathbf{h} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Note: If the class convention is correlation (no flip), the 1 goes at row 2, col 4 instead.

(c) **[4 points]** Let $\mathbf{X}$ denote the Fourier transform of the image $\mathbf{x} \in \mathbb{R}^{K \times K}$, let $\mathbf{H}$ denote the Fourier Transform of the filter $\mathbf{h}$. Let $\mathbf{Y} \triangleq \mathbf{X} \odot \mathbf{H}$ be the Fourier transform of $\mathbf{y}$, where $\odot$ denotes the element-wise multiplication.

The frequency response $\mathbf{H}$ of the filter $\mathbf{h}$ is shown in Figure 1 of the exam: $\mathbf{H}$ is nonzero *only* at the DC component $(0,0)$, where $H[0,0] = 1$, and zero everywhere else.

Given the $\mathbf{H}$ in Figure 1, write $\mathbf{y}$ in terms of $\mathbf{x}$ (without using $\mathbf{H}$ or $\mathbf{h}$). Explain how you came up with the solution.

Solution

$\mathbf{H}$ is nonzero *only* at the DC component $(0,0)$, where $H[0,0] = 1$, and $H[u,v] = 0$ for all other frequencies. Therefore $\mathbf{Y} = \mathbf{X} \odot \mathbf{H}$ retains only the DC term:

$$Y[0,0] = X[0,0], \qquad Y[u,v] = 0 \text{ for } (u,v) \neq (0,0).$$

Since $X[0,0] = \sum_{i,j} x[i,j]$, the inverse DFT of $\mathbf{Y}$ gives a **constant image**:

$$y[i,j] = \frac{1}{K^2} X[0,0] = \frac{1}{K^2} \sum_{i,j} x[i,j] = \bar{x} \qquad \forall\, i,j$$

$$\boxed{\mathbf{y} = \frac{1}{K^2} \left(\sum_{i,j} x[i,j] \right) \cdot \mathbf{1} = \bar{x} \cdot \mathbf{1}}$$

That is, every pixel of $\mathbf{y}$ equals the **mean** of $\mathbf{x}$ (the sum of all pixels normalized by K^2).

Reasoning: H is a delta at DC $\Rightarrow$ only the zeroth frequency survives $\Rightarrow$ the output is the pixel average broadcast to every location.

3. [10 points] Fitting & Alignment

(a) **[4 points]** In image alignment, instead of a full affine transformation, we wish to fit a **similarity transformation** (rotation + uniform scale + translation):

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s\cos\theta & -s\sin\theta \\ s\sin\theta & s\cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}.$$

Let $a = s\cos\theta$ and $b = s\sin\theta$.

Given $N = 3$ keypoint pairs $\{(\mathbf{x}_i, \mathbf{x}_i')\}_{i=1}^N$, formulate the least-squares problem

$$\min_{\mathbf{v}} \|\mathbf{A}\mathbf{v} - \mathbf{c}\|^2$$

by defining $\mathbf{v}$, $\mathbf{A}$, and $\mathbf{c}$ (with their shapes) for $N = 3$ pairs.

Solution

With $a = s\cos\theta$, $b = s\sin\theta$, each correspondence gives two equations:

$$x_i' = a\,x_i - b\,y_i + t_x$$
$$y_i' = b\,x_i + a\,y_i + t_y$$

Define $\mathbf{v} = \begin{bmatrix} a & b & t_x & t_y \end{bmatrix}^\top \in \mathbb{R}^4$.

For $N = 3$ pairs:

$$\mathbf{A} = \begin{bmatrix} x_1 & -y_1 & 1 & 0 \\ y_1 & x_1 & 0 & 1 \\ x_2 & -y_2 & 1 & 0 \\ y_2 & x_2 & 0 & 1 \\ x_3 & -y_3 & 1 & 0 \\ y_3 & x_3 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{6\times 4}, \qquad \mathbf{c} = \begin{bmatrix} x_1' \\ y_1' \\ x_2' \\ y_2' \\ x_3' \\ y_3' \end{bmatrix} \in \mathbb{R}^6$$

The least-squares solution is $\mathbf{v}^\star = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{c}$.

(b) **[4 points]** To estimate a homography, we need at least four pairs of corresponding 2D points, because there are 8 free parameters.

(i) Suppose you have 8 pairs of corresponding points, and you are confident that they are all roughly correct but with a small measurement error. How should you use them to estimate the homography (e.g., should you use all eight pairs or just four; describe the method)?

Solution

Use **all 8 pairs** with the **Direct Linear Transform (DLT)**. Set up the overdetermined 16×9 linear system $\mathbf{Ah} = \mathbf{0}$ (2 equations per pair) and solve via SVD: the solution is the right singular vector corresponding to the smallest singular value.

Using all 8 pairs gives a least-squares fit that averages out the small measurement errors, yielding a more robust estimate than using only 4.

(ii) Now, suppose that you have 8 pairs of corresponding points, but some of them might be outliers (i.e., not even close to correct). How should you use them to estimate the homography (e.g., should you use all 8 pairs; describe the method)?

Solution

Use **RANSAC**:

i. Randomly sample 4 pairs (the minimum needed for a homography).
ii. Compute the homography from those 4 pairs (via DLT).
iii. Count inliers among all 8 pairs (points whose reprojection error is below a threshold ϵ).
iv. Repeat for many iterations; keep the model with the most inliers.
v. (Optional) Refit using all inliers of the best model via DLT.

RANSAC is robust to outliers because each hypothesis is computed from a minimal sample, and outliers are unlikely to all appear in the same random draw. *Note*: even saying that they will use RANSAC because it is robust to outliers it is OK. They need to say that they will use 4 pairs which is the minimum number of samples required to solve for the transformation.

(c) **[2 points]** Let λ_1 and λ_2 be the eigenvalues of the second-moment matrix at a pixel. Suppose we define the corner score as $\tilde{R} = \min(\lambda_1, \lambda_2)$. Using only $\tilde{R}$, can you distinguish flat regions, edges, and corners? Please explain.

Solution

$\tilde{R} = \min(\lambda_1, \lambda_2)$ can identify **corners** (both eigenvalues large $\Rightarrow \tilde{R}$ large), but it **cannot distinguish flat regions from edges**:

- **Flat:** both $\lambda_1, \lambda_2 \approx 0 \Rightarrow \tilde{R} \approx 0$.
- **Edge:** one large, one $\approx 0 \Rightarrow \tilde{R} \approx 0$.
- **Corner:** both large $\Rightarrow \tilde{R}$ is large.

Since both flat and edge give $\tilde{R} \approx 0$, $\tilde{R}$ alone cannot separate them. One would additionally need $\max(\lambda_1, \lambda_2)$ or the full Harris score $\det - k\,\mathrm{trace}^2$ to distinguish edges from flat regions.

4. [22 points] Short Answer

(a) **[12 points]** Below are 6 images and their corresponding Fourier Transform (magnitude only). Please match them **and** provide a very short explanation of how you are matching them.

Images:

(1) Wide Gaussian (2) Brick wall (3) Diagonal Gabor (4) Dipole (5) Noise (6) Horiz. line

Fourier Transforms: (a) through (f) as shown in the exam.

Solution

Image	FT	Explanation
(1)	**(c)**	*Gaussian blob.* A wide, smooth Gaussian in the spatial domain transforms to a narrow, concentrated Gaussian at the origin in the frequency domain. FT (c) shows a compact bright spot.
(2)	**(f)**	*Brick wall.* The periodic rectangular grid produces regularly spaced impulses in the frequency domain. FT (f) shows a grid-like pattern of periodic peaks.
(3)	**(e)**	*Diagonal sinusoidal grating / Gabor.* A sinusoidal grating produces exactly two symmetric impulses oriented perpendicular to the grating stripes. FT (e) shows two symmetric dots. Another way to answer this is to observe that 3 is the multiplication of a sinusoidal grating with a Gaussian and hence the Fourier of this is the convolution of the Fourier of the sinusoid with the Gaussian, thus two blobs spaced diagonally.
(4)	**(b)**	*Two point sources (dipole).* The FT of two delta-like points is a sinusoidal fringe pattern oriented along the axis connecting the points. FT (b) shows oriented fringes. An alterative way is to observe that the Fourier of 4 is image 3 based on Fourier duality.
(5)	**(a)**	*White noise.* White noise has a flat (uniform) power spectrum. FT (a) shows roughly uniform, noisy energy spread across all frequencies.
(6)	**(d)**	*Horizontal edge / step.* A horizontal step edge concentrates energy along the vertical frequency axis. FT (d) shows a vertical line through the center.

One point for correct match one point for correct reasoning.

(b) **[10 points]** Below are 5 images and their corresponding Hough Transform parameterized by (ρ, θ), where $\rho = x\cos\theta + y\sin\theta$. Each point (x, y) in image space maps to a sinusoidal curve in (ρ, θ) space. Please match them **and** provide a very short explanation of how you are matching them.

Images:

(1) Two lines (2) Three lines (3) V shape (4) Symmetric X (5) Rectangle

Hough Transforms: (a) through (e) as shown in the exam.

Solution

In the Hough transform with $\rho = x\cos\theta + y\sin\theta$, each image point traces a sinusoidal curve in (ρ, θ) space. A line in the image corresponds to an intersection point (peak) where many sinusoids meet.

Image	**HT**	**Explanation**
(1)	**(d)**	One line one sinusoidal-like pattern with one peak.
(2)	**(a)**	Two parallel lines correspond to two curves with two unique peaks. Since lines are parallel they have the same parameter θ and different ρ, thus the peaks will be at the same x location.
(3)	**(c)**	Two near-vertical lines (V shape) correspond to θ values near 0 or π. HT (c) shows two intersections at similar θ.
(4)	**(e)**	A symmetric X pattern (two lines crossing at the center) gives two well-separated, symmetric peaks. HT (d) shows this symmetric structure.
(5)	**(b)**	A rectangle has four lines (two pairs of parallels). HT (b) shows four peaks, with pairs sharing the same θ but opposite ρ.

Note: Exact matching depends on the precise geometry visible in the printed images. Credit is awarded for consistent reasoning.

5. [8 points] Multiple Choice

Please write the letter in the answer box. Only one answer is correct.

1. [**1 point**] Which of the following best describes the "aperture problem" in optical flow?

(a) The optical flow is ambiguous because camera aperture affects depth of field.
(b) Through a small local window, only the component of motion perpendicular to an edge can be determined.
(c) Large motions exceed the linear approximation used in gradient-based methods.
(d) The brightness constancy assumption fails for specular surfaces.

Solution

(b) — Through a small local window, only the component of motion perpendicular to an edge can be determined.

2. [**1 point**] When applying a Hough transform, noise can be countered by

(a) a finer discretization of the accumulator
(b) increasing the threshold on the number of votes a valid model has to obtain
(c) decreasing the threshold on the number of votes a valid model has to obtain
(d) considering only a random subset of the points since these might be inliers

Solution

(b) — Increasing the vote threshold filters out spurious peaks caused by noise.

3. [**1 point**] SIFT is designed to detect and describe keypoints invariant to scale changes. Conceptually, what enables SIFT to identify the same physical feature even when an image is resized?

(a) It computes features using a histogram of gradient orientations relative to each keypoint's dominant (most common) direction.
(b) It searches for local extrema in a scale-space representation.
(c) It normalizes keypoint descriptors using bilinear interpolation to make the features independent of image scale.
(d) It uses affine transformations to align features across different image scales, which builds a consistent representation.

Solution

(b) — SIFT achieves scale invariance by searching for local extrema in a **scale-space representation** (Difference-of-Gaussians pyramid).

4. [**1 point**] Which of the following transformations preserves parallelism of lines but not angles?

(a) Projective (homography)
(b) Affine
(c) Euclidean (rigid)
(d) Radial distortion

Solution

(b) — **Affine** transformations preserve parallelism but not angles. (Euclidean preserves both; projective preserves neither.)

5. [**1 point**] The Fourier transform of a 2D Gaussian is:

(a) Another Gaussian
(b) A delta function
(c) A sinusoid
(d) A step function

Solution

(a) — The Fourier transform of a Gaussian is **another Gaussian**.

6. [**1 point**] The main advantage of using Gaussian smoothing before edge detection is

(a) It removes noise that may cause false edges.
(b) It normalizes pixel intensity.
(c) It enhances sharp edges.
(d) It reduces the image range.

Solution

(a) — Gaussian smoothing **removes noise** that may cause false edges.

7. [**1 point**] Which of the following will most significantly increase the number of iterations required by RANSAC?

(a) Increasing the total number of data points
(b) Increasing the percentage of outliers
(c) Using a larger inlier threshold
(d) Increasing the image resolution

Solution

(b) — The required RANSAC iterations scale as $(1 - w^n)^{-1}$ where w is the inlier ratio. Increasing the outlier percentage (decreasing w) dramatically increases iterations.

8. **[1 point]** A thin lens has focal length f. An object is placed exactly at distance $S = f$ from the lens. Where does the image form?

(a) At distance $S' = f$

(b) At distance $S' = 2f$

(c) At infinity

(d) No image is formed

Solution

(c) — From the thin lens equation $\frac{1}{S} + \frac{1}{S'} = \frac{1}{f}$: if $S = f$, then $\frac{1}{S'} = 0$, so $S' = \infty$. The image forms **at infinity**.

End of Exam — Questions & Solutions